

EXERCISE 3.5

Solve the following.

(i) $\frac{4}{9} \div \frac{16}{9}$

(ii) $\frac{4}{8} \div \frac{2}{12}$

(iii) $\frac{15}{20} \div \frac{3}{12}$

(iv) $\frac{1}{4} \div \frac{7}{16}$

(v) $\frac{9}{10} \div \frac{12}{15}$

(vi) $\frac{9}{30} \div \frac{6}{12}$

(vii) $\frac{12}{21} \div \frac{6}{27}$

(viii) $\frac{5}{25} \div \frac{20}{30}$

(ix) $2\frac{1}{3} \div \frac{2}{5}$

(x) $2\frac{1}{3} \div \frac{2}{4}$

(xi) $2\frac{4}{5} \div \frac{2}{3}$

(xii) $3\frac{6}{83} \div 4\frac{2}{4}$

(xiii) $6 \div \frac{2}{3}$

(xiv) $\frac{2}{3} \div 8$

(xv) $\frac{12}{5} \div 9$

Solve real life problems involving fractions using all four operations

Example 1: One jar contains $\frac{1}{2}$ kg of sugar. Another jar contains $\frac{1}{4}$ kg of sugar. What is the total quantity of the sugar?

Solution: $\frac{1}{2} + \frac{1}{4} = \frac{2+1}{4} = \frac{3}{4}$

Thus, the total quantity of the sugar is $\frac{3}{4}$ kg.

Teacher's Note

Teacher should teach the students how to change real life problems in mathematical form and solve them.

Example 2: Nasima purchased $4\frac{1}{4}$ kg of milk powder. She used $1\frac{3}{8}$ kg of it in a month. How much milk powder is left?

Solution: Quantity of milk powder = $4\frac{1}{4}$ Kg.

Quantity of milk powder used = $1\frac{3}{8}$ Kg.

Subtract milk powder used from total milk powder

$$\begin{aligned}\text{Quantity of milk powder left} &= 4\frac{1}{4} - 1\frac{3}{8} \\ &= \frac{17}{4} - \frac{11}{8} \\ &= \frac{34}{8} - \frac{11}{8} = \frac{34-11}{8} = \frac{23}{8} \\ &= 2\frac{7}{8}\end{aligned}$$

Hence, the remaining milk powder left is $2\frac{7}{8}$

Example 3: What will be the total length of 5 pieces of string, if each piece is $\frac{3}{4}$ metre long.

Solution: Length of 1 piece = $\frac{3}{4}$ m

$$\text{Length of 5 pieces} = \frac{3}{4} \times 5 = \frac{15}{4} = 3\frac{3}{4} \text{ m}$$

Thus the total length of 5 pieces of string is $3\frac{3}{4}$ m

EXERCISE 3.6

1. A family used $4\frac{1}{4}$ litres of milk and another family used $7\frac{1}{2}$ litres of milk in a day. How much quantity of milk was used by both families in the day?
2. My aunty buys a piece of ribbon that was $4\frac{1}{6}$ cm long. She buys another piece of ribbon that was $4\frac{2}{5}$ cm long. How much longer the second piece of ribbon was?
3. I am $8\frac{1}{2}$ years old and my brother is $2\frac{1}{2}$ years younger than me, what is the age of my brother?

4. Nazia has $4\frac{4}{5}$ metres of ribbon. She used $\frac{1}{2}$ of this ribbon to tie a present for her daughter. How many metres of ribbon did she use?
5. A tree is 18 metre tall. How many pieces of wood can be cut from the tree, if each piece is to be $\frac{2}{3}$ metre long?
6. Shopkeeper has 15 kg of peanuts. He places $\frac{2}{5}$ kgs of peanuts in a bag. How many bags will he use?

REVIEW EXERCISE – 3

1. Convert the following into mixed numbers:

(i) $\frac{66}{8}$ (ii) $\frac{17}{3}$ (iii) $\frac{23}{5}$ (iv) $\frac{40}{5}$

2. Convert the following mixed fractions into improper fractions:

(i) $8\frac{4}{5}$ (ii) $3\frac{5}{8}$ (iii) $7\frac{3}{7}$ (iv) $4\frac{3}{2}$

3. Write down the following numbers in ascending order:

(i) $\frac{3}{8}, \frac{9}{8}, \frac{5}{8}$ (ii) $\frac{2}{3}, \frac{1}{4}, \frac{5}{6}$
(iii) $\frac{6}{4}, \frac{8}{3}, \frac{7}{6}, \frac{5}{2}$ (iv) $\frac{2}{7}, \frac{5}{11}, \frac{4}{5}, \frac{3}{11}$

4. Write down the following fractions in descending order:

(i) $\frac{13}{6}, \frac{11}{6}, \frac{7}{6}$ (ii) $\frac{2}{3}, \frac{3}{5}, \frac{5}{6}$
(iii) $\frac{4}{15}, \frac{3}{10}, \frac{9}{20}, \frac{4}{5}$ (iv) $\frac{4}{3}, \frac{3}{5}, \frac{5}{7}, \frac{1}{3}$

5. Reduce the following fractions into lowest terms:

(i) $\frac{30}{42}$ (ii) $\frac{24}{44}$ (iii) $2\frac{4}{8}$ (iv) $3\frac{3}{6}$

6. Simplify:

(i) $\frac{3}{5} + \frac{2}{4}$

(ii) $\frac{7}{9} + \frac{4}{6} + \frac{2}{3}$

(iii) $\frac{4}{5} + \frac{6}{10} + \frac{1}{2}$

(iv) $\frac{2}{3} + 4\frac{1}{2}$

7. Simplify:

(i) $\frac{3}{4} - \frac{4}{8}$

(ii) $\frac{11}{5} - \frac{9}{5}$

(iii) $5\frac{2}{3} - 4\frac{2}{4}$

(iv) $\frac{13}{15} - \frac{5}{10} - \frac{1}{5}$

8. Multiply the following fractions and give your answer in lowest terms:

(i) $2\frac{2}{5} \times 2\frac{1}{2}$

(ii) $1\frac{2}{3} \times 2\frac{1}{4}$

(iii) $\frac{28}{35} \times \frac{12}{21}$

(iv) $1\frac{3}{5} \times 1\frac{5}{10}$

9. Divide the following fractions and give your answer in lowest terms:

(i) $\frac{5}{6} \div 25$

(ii) $\frac{3}{4} \div \frac{5}{8}$

(iii) $5\frac{1}{2} \div 1\frac{1}{4}$

(iv) $7\frac{3}{5} \div 1\frac{7}{12}$

- 10.** Nasir and his friend were jogging on a track. Nasir jogged $7\frac{1}{2}$ km and his friend jogged $4\frac{2}{3}$ km. How much more distance did Nasir cover than his friend?

- 11.** There are 32 balloons in a room, out of which $\frac{1}{4}$ are red. How many balloons in the room are red?

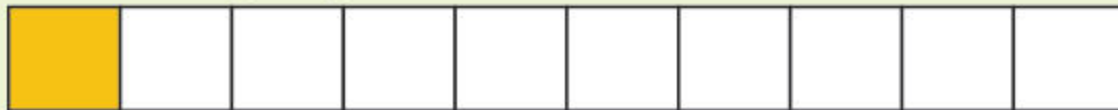
- 12.** Javeria's doll dress requires $\frac{3}{4}$ metre of fabric. How many dresses of doll can be made from $3\frac{3}{4}$ metres of fabric?

- 13.** A factory makes $9\frac{1}{2}$ litres of apple juice each hour. How many litres of apple juice will the factory make in 14 hours?

In previous unit, we have learnt about fractions.

Define decimal as a fraction whose denominator is 10 or power of 10

Look at the figure given below.



In **fraction form** the shaded portion is written as

$\frac{1}{10}$ and read as one tenth.

There will be one digit after the decimal point if the whole is divided into 10 equal parts

Whole number

0.1

Fractional part

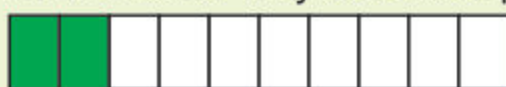
Decimal point

Here dot (.) is known as the decimal point.

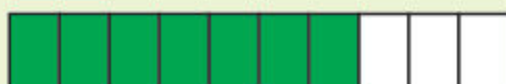
It separates the whole number part and the fractional part.

Unit 4 DECIMALS AND FRACTIONS (Decimals)

In the same way coloured portion of:



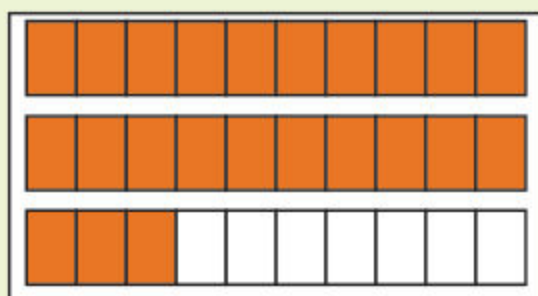
represents $\frac{2}{10}$ or 0.2



represents $\frac{7}{10}$ or 0.7



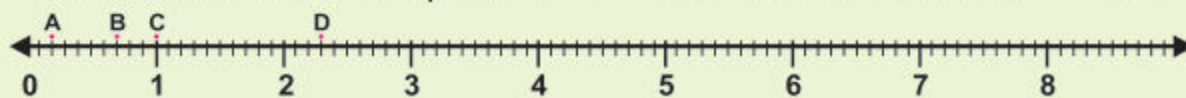
represents $\frac{10}{10} = 1.0$ or (1 whole)



represents $2\frac{3}{10}$ or 2.3

(2 whole and 3 tenths)

All this can also be represented on a number line shown below.



Point A represents $\frac{2}{10}$ or 0.2, point B represents $\frac{7}{10}$ or 0.7,

point C represents $\frac{10}{10}$ or 1.0 and point D represents $2\frac{3}{10}$ or 2.3.

Each of the following figure is divided into 100 equal parts. The coloured portion of these figures represents:

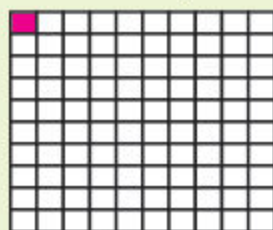


Figure 1

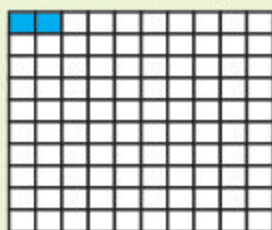


Figure 2

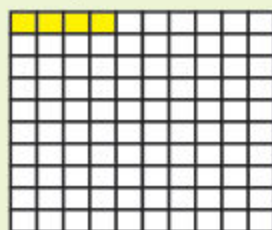


Figure 3

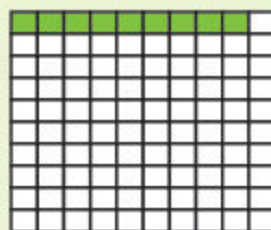


Figure 4

Figure 1: One-hundredth = $\frac{1}{100} = 0.01$ (Read as zero point zero one.)

Figure 2: Two-hundredth = $\frac{2}{100} = 0.02$ (Read as zero point zero two.)

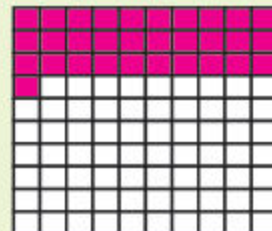
Figure 3: Four-hundredth = $\frac{4}{100} = 0.04$ (Read as zero point zero four.)

Figure 4: Nine-hundredth = $\frac{9}{100} = 0.09$ (Read as zero point zero nine.)

Unit 4 DECIMALS AND FRACTIONS (Decimals)

Example 1: The whole is divided into 100 equal parts, and 31 of them are shaded. Write in decimal fraction.

In fraction form it is written as $\frac{31}{100}$ and read as thirty one hundredths. Where as, in decimal form it is written as 0.31

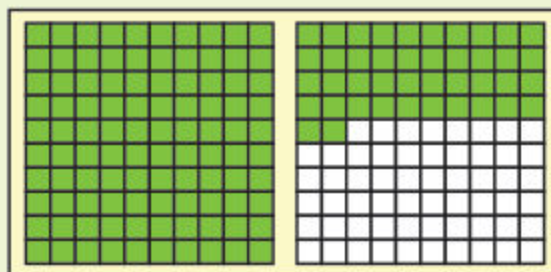


Note that there are 2-digits after the decimal point, if the whole is divided into 100 equal parts.

0.31
and read as
“zero point
three one”

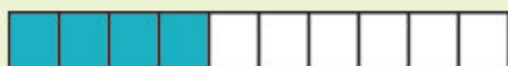
Similarly, in the given figure
coloured portion $1\frac{42}{100}$
represents or 1.42

Read as one point four two.



Activity

Write the fractions as well as the decimal fractions represented by the coloured portions of the figures:



Fraction = $\frac{4}{10}$

Decimal fraction = 0.4



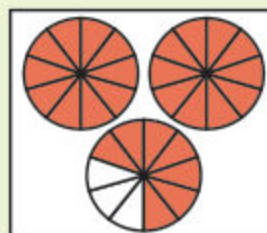
Fraction =

Decimal fraction =



Fraction =

Decimal fraction =



Fraction =

Decimal fraction =

Unit 4 DECIMALS AND FRACTIONS (Decimals)

Recognize the places occupied by the digits, after the decimal point, as decimal places

The number of digits after the decimal point are called the **Decimal Places**.

Look at the following examples.

- (1) 3.2 represents decimal fraction up to **one decimal place** as it has only one digit after the decimal point.

Fractional part has a denominator of 10.

- (2) 1.43 represents decimal fraction up to **two decimal places** as it has two digits after the decimal point.

Fractional part has a denominator of 100.

- (3) 1.324 represent decimal fraction up to **three decimal places** as it has three digits after the decimal point.

Fractional part has a denominator of 1000.



Activity

Fill the given table with the denominator and the decimal places for the decimal numbers given below:

Decimals	Denominator	No. of decimal places
0.3	10	One
13.5		
0.84		
732.52		
0.324		
583.123		

Unit 4 DECIMALS AND FRACTIONS (Decimals)

Identify the place value of a digit in decimals

In decimal fraction, the value of the digit after the decimal point decreases by 10 times.

Look at the following table to understand the place values in decimal fraction.

Fractions	H T O	.	t tenth	h hundredth	th thousandth
$\frac{1}{10}$	0	.	1		
$\frac{1}{100}$	0	.	0	1	
$\frac{1}{1000}$	0	.	0	0	1

- The first digit after the decimal point has a place value of 'tenths' and it is the first decimal place.
- The second digit after the decimal point has a place value equal to 'hundredths' and it is the second decimal place.
- The third digit after the decimal point has a place value equal to 'thousandths' and it is the third decimal place.

Example 1: Write the place values and value of bold digits in the following numbers:

Number	Place Value	Value
53. 5 79	Tenths	5 tenths or 0.5 or $\frac{5}{10}$
1.0 3 2	Hundredths	3 hundredths or 0.03 or $\frac{3}{100}$
123.87 3	Thousandths	3 thousandths or 0.003 or $\frac{3}{1000}$

Example 2: Identify the place value and value of each digit in 23.416.

Let's first write the given number under the place value chart.

T	O	.	T	H	TH
2	3	.	4	1	6

2 tens or 20 3 ones or 3 4 tenths or 0.4 1 hundredths or 0.01 6 thousandths or 0.006

Place value of 2 = 2 tens = $2 \times 10 = 20$

Place value of 3 = 3 ones = $3 \times 1 = 3$

Place value of 4 = 4 tenths = $\frac{4}{10} = 0.4$

Place value of 1 = 1 hundredths = $\frac{1}{100} = 0.01$

Place value of 6 = 6 thousandths = $\frac{6}{1000} = 0.006$

EXERCISE 4.1

1. Draw the figures for the following decimal fractions.

(i) 0.3

(ii) 0.8

(iii) 1.3

(iv) 1.7

(v) 0.34

(vi) 1.5

2. Write down the number of decimal places in each.

(i) 123.1

(ii) 57.322

(iii) 0.87

(iv) 139.45

(v) 59.552

(vi) 735.9

3. Write down the place value and value of encircled digit.

(i) 725.0 (4)

(ii) 135. (8) 5

(iii) 5.09 (2)

(iv) (1) 24.3 (2)

(v) (4) 3.5 (9)

(vi) (3) 4.6 (7)

4. Write down the place value and value of each digit.

(i) 575.1

(ii) 0.534

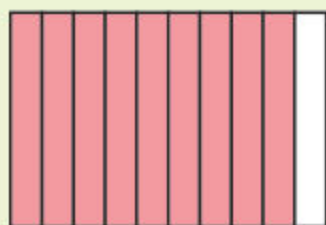
(iii) 18.97

(iv) 9.02

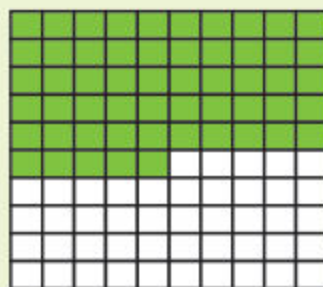
(v) 75.84

(vi) 51.15

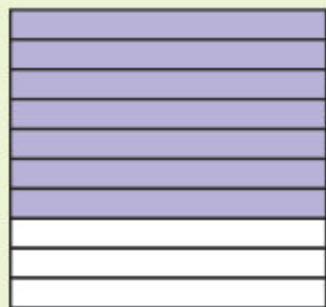
5. Match the coloured part of figures with the correct decimal fractions.



1.4



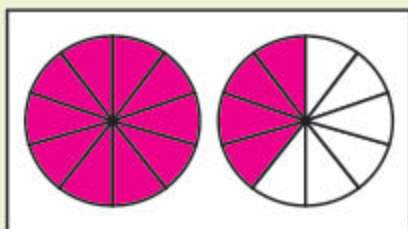
0.23



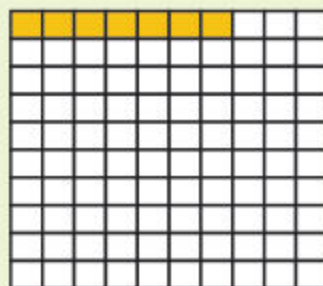
0.07



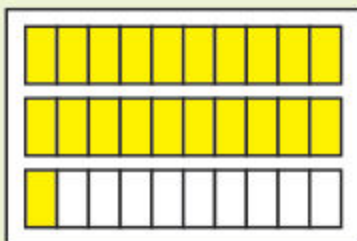
2.1



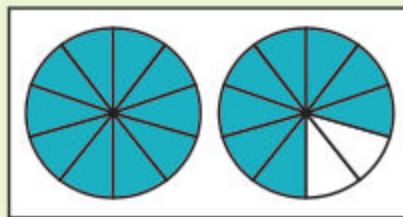
1.8



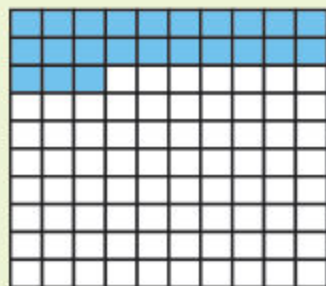
2.2



0.9



0.55



1.6



0.7

4.2 CONVERSION BETWEEN FRACTIONS AND DECIMALS

Convert a given fraction to a decimal if,
(i) denominator of the fraction is 10 or a power of 10.

In order to understand how to convert the fraction into decimal when the denominator is 10, 100 or 1000.

Consider the following examples.

Example 1: Convert $\frac{3}{10}$ into decimal.

Solution: $\frac{3}{10} = 3$ tenths.

So, $\frac{3}{10} = 0.3$

Example 2: Convert $\frac{3}{100}$ into decimal.

Solution: Here, $\frac{3}{100} = 3$ hundredths.

So, $\frac{3}{100} = 0.03$ because there are no tenths.

But, if we have to convert $\frac{13}{100}$ into decimal, it will be 0.13

Example 3: Convert $\frac{23}{10}$ into decimal.

Solution: Since, it is an improper fraction, it is easier to first convert it into mixed fraction.

$$\frac{23}{10} = 2 \frac{3}{10}$$

Now $2 \frac{3}{10}$ means 2 wholes and 3 tenths, i.e. 2.3

So, $\frac{23}{10} = 2.3 \rightarrow$ fraction

↓

whole number

So, in 2.3, 2 is whole number and .3 or $\frac{3}{10}$ is fraction.

Example 4: Convert $\frac{32}{1000}$ into decimal.

Solution: $\frac{32}{1000}$ means 32 thousandths.

$$\text{So, } \frac{32}{1000} = 0.032$$

(ii) Conversion of fractions into decimals when the denominator is not 10, 100 or 1000

Example 1: Convert $\frac{1}{2}$ into decimal.

Solution: Here, the denominator is not 10 or a power of 10 but it can be converted into 10 or powers of 10.

$$\text{Now } \frac{1}{2} = \frac{1}{2} \times \frac{5}{5}$$

▪ **Making denominator multiple of 10 or power of 10**

$$= \frac{5}{10} \text{ is equivalent to } \frac{1}{2}$$

$$= 5 \text{ tenths} = 0.5, \text{ this means } \frac{1}{2} = 0.5$$

Example 2: Convert $\frac{13}{4}$ into decimal.

$$\text{Solution: } \frac{13}{4} = 3\frac{1}{4} \quad [\text{mixed number}]$$

$$= 3 + \frac{1}{4} \quad [3 \text{ wholes and one fourth}]$$

$$= 3 + \frac{1}{4} \times \frac{25}{25} \quad \text{▪ } \frac{1}{4} \text{ can be converted into its equivalent fraction with denominator 100}$$

$$= 3 + \frac{25}{100}$$

$$= 3 + \frac{25}{100}$$

$$= 3.25$$

▪ **3 wholes and 25 hundredths**

Example 3: Convert $\frac{11}{25}$ into decimal.

Solution:

$$\begin{aligned}\frac{11}{25} &= \frac{11}{25} \times \frac{4}{4} \\ &= \frac{44}{100} = 0.44\end{aligned}$$

Convert decimals (up to three decimal places) to fractions

Let's consider the following examples.

Example 1: Convert 0.4 into fraction.

Solution: 0.4 = 4 tenths

$$\begin{aligned}&= \frac{4}{10} && \bullet \text{ 1}^{\text{st}} \text{ decimal place is tenths;} \\ &&& \text{so, denominator will be 10} \\ &= \frac{\overset{2}{\cancel{4}}}{\underset{5}{\cancel{10}}} = \frac{2}{5}\end{aligned}$$

Example 2: Convert 0.32 into fraction.

Solution: 0.32 = 32 hundredths

$$\begin{aligned}&= \frac{32}{100} && \bullet \text{ 2}^{\text{nd}} \text{ decimal place hundredth:} \\ &&& \text{So, denominator will be 100} \\ &= \frac{\overset{16}{\cancel{32}}}{\underset{50}{\cancel{100}}} && \bullet \text{ Simplifying dividing by 4} \\ &= \frac{\overset{8}{\cancel{16}}}{\underset{25}{\cancel{50}}} \\ &= \frac{8}{25}\end{aligned}$$

Example 3: Convert 1.24 into fraction.**Solution:** $1.24 = 1$ whole and 24 hundredths

$$\begin{aligned}
 1.24 &= 1 \frac{24}{100} \\
 &= 1 \frac{\overset{6}{\cancel{24}}}{\underset{25}{\cancel{100}}} = 1 \frac{6}{25} \quad \text{Dividing by 4}
 \end{aligned}$$

Example 4: Convert 0.135 into fraction.**Solution:** $0.135 = 135$ thousandths

$$\begin{aligned}
 0.135 &= \frac{135}{1000} \\
 &= \frac{\overset{27}{\cancel{135}}}{\underset{200}{\cancel{1000}}} = \frac{27}{200} \quad \text{Dividing by 5}
 \end{aligned}$$

EXERCISE 4.2**1. Convert the following fractions into decimals.**

(i) $\frac{1}{100}$	(ii) $\frac{5}{10}$	(iii) $\frac{23}{100}$	(iv) $\frac{431}{1000}$
(v) $\frac{231}{100}$	(vi) $\frac{17}{10}$	(vii) $\frac{29}{1000}$	(viii) $\frac{11}{100}$

2. Convert the following fractions into decimals.

(i) $\frac{7}{25}$	(ii) $\frac{42}{4}$	(iii) $\frac{9}{2}$	(iv) $\frac{3}{4}$
(v) $\frac{13}{50}$	(vi) $\frac{3}{8}$	(vii) $\frac{19}{25}$	(viii) $\frac{30}{50}$

3. Convert the following decimals into fractions and give your answer in simplest form.

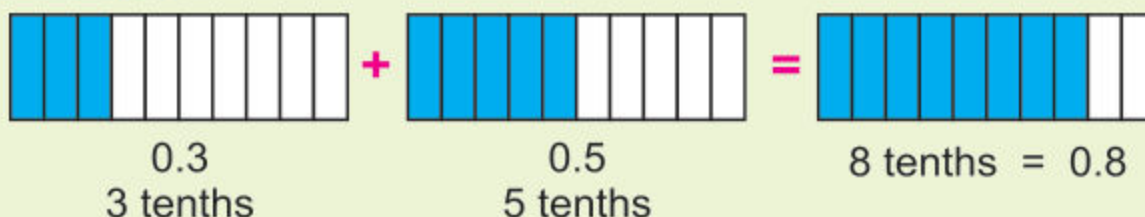
(i) 0.3	(ii) 1.35	(iii) 12.37	(iv) 0.432
(v) 25.5	(vi) 2.32	(vii) 0.45	(viii) 135.2

4.3 BASIC OPERATION ON DECIMALS

Add and subtract decimals (up to three decimal places)

Look at the following pictorial example to understand addition of decimals

Example 1:



We can write it as:

$$\begin{array}{r}
 0.3 \\
 + 0.5 \\
 \hline
 0.8
 \end{array}$$

- Align the decimal point, i.e put decimal point just one below the other.
- Start from the digit at the lowest place value.
- Do the usual addition.
- Decimal point in the answer also should be aligned.

Example 2: Subtract.



$$\begin{array}{r}
 0.6 - 0.2 = 0.4 \\
 6 \text{ tenths} - 2 \text{ tenths} = 4 \text{ tenths}
 \end{array}$$

$$\begin{array}{r}
 0.6 \\
 - 0.2 \\
 \hline
 0.4
 \end{array}$$

Example 3: Subtract the following.

(i) $3.57 - 2.43$

$$\begin{array}{r}
 3.57 \\
 - 2.43 \\
 \hline
 1.14
 \end{array}$$

(ii) $7.84 - 1.75$

$$\begin{array}{r}
 7.84 \\
 - 1.75 \\
 \hline
 6.09
 \end{array}$$

EXERCISE 4.3

1. Add the following.

(i) $0.35 + 0.42$

(iii) $1.27 + 3.49$

(v) $1.34 + 0.43$

(vii) $0.3 + 0.9 + 0.4$

(ix) $68.378 + 12.633$

(ii) $0.21 + 0.35 + 0.11$

(iv) $13.5 + 14.2 + 7.9$

(vi) $25.32 + 15.75$

(viii) $1.5 + 3.9 + 17.2$

(x) $42.243 + 81.483$

2. Subtract the following.

(i) $0.9 - 0.3$

(iii) $10.59 - 3.27$

(v) $15.39 - 12.41$

(vii) $19.31 - 10.32$

(ix) $49.243 - 17.157$

(ii) $0.84 - 0.61$

(iv) $13.54 - 12.87$

(vi) $87.34 - 85.56$

(viii) $54.79 - 39.84$

(x) $76.919 - 64.883$

Multiply decimal by 10, 100 and 1000

Multiplication of decimals by 10, 100 or 1,000 just invites moving the decimal point to the right depending upon the power of 10 we are multiplying with.

Example 1: Multiply 3.24×10

Solution: 3.24×10

$$= 32.4$$

Here decimal number 3.24 has multiplied by 10.
The number has moved one column / place to left.

$$\begin{array}{r} 3.24 \\ \times 10 \\ \hline 0000 \\ 324x \\ \hline 32.40 \end{array}$$
 (We multiply numbers where decimal fraction is of two place value. Count digits from right to left. Put decimal after two numbers)

Here the decimal point moves one place to the right.

Example 2:

Multiply 0.325 by 100

$$\begin{array}{r} 0.325 \\ \times 100 \\ \hline 32.5 \end{array}$$

Decimal point moves two places to the right.

Example 3:

Multiply 1.834 x 1000

$$\begin{array}{r} 1.834 \\ \times 1000 \\ \hline 1834.0 \\ \text{Or } 1834 \end{array}$$

Decimal point in the end has no value, no fractional part.

Decimal point moves three places to the right.

Multiply a decimal by a 2-digit number

Look at the following examples.

Example 1: Multiply 13.5 x 15

Solution:

$$\begin{array}{r} 13.5 \\ \times 15 \\ \hline 675 \\ 1350 \\ \hline 202.5 \end{array}$$

- Do the multiplication just as in the case of whole numbers, ignoring the decimal point for a while.
- Multiply the units or ones
- Multiply the tens
- Now look at the total number of decimal places in the numbers being multiplied. There is only one decimal place so, the product should also have one decimal place. Put decimal at one decimal place from right.
- One decimal place In the product

Hence, $13.5 \times 15 = 202.5$

EXERCISE 4.4

Do the following multiplications.

- | | | |
|-------------------------|--------------------------|-------------------------|
| (1) 6.3×10 | (2) 0.25×10 | (3) 1.732×100 |
| (4) 0.327×100 | (5) 22.3542×100 | (6) 1.265×1000 |
| (7) 1.872×1000 | (8) 6.3×3 | (9) 0.85×25 |
| (10) 7.9×13 | (11) 7.21×33 | (12) 1.834×11 |
| (13) 5.42×24 | (14) 0.49×71 | (15) 3.52×45 |

Divide a decimal by a 1-digit number (quotient being a decimal up to two decimal places)

To divide a decimal number by a one digit number (whole number), follow the usual method of dividing whole numbers, just remember to place the decimal point in the quotient when it comes in the dividend.

Look at the following examples to understand this concept.

Example : Divide **27.5** by **5**

Solution:

$$\begin{array}{r}
 5.5 \\
 5 \overline{) 27.5} \\
 \underline{25} \\
 25 \\
 \underline{25} \\
 00
 \end{array}$$

[$5 \times 5 = 25$]

[$5 \times 5 = 25$ again]

Steps:

- Start from whole numbers we have 27. Divide 27 by 5.
- Because $5 \times 5 = 25$
- Now 2 is smaller than the divisor 5, so bring next 5 down, but before that there is a decimal point so shift it to the quotient.

Hence, $27.5 \div 5 = 5.5$

Teacher's Note

Teacher should teach the students to divide decimal by 1-digit numbers and give them more examples to solve for sufficient practice.